
VAIBHAV SHARMA

DATA SCIENTIST

New Delhi, India 110025 ♦ 8800707522 ♦ svaibhav9720@gmail.com

PROFESSIONAL SUMMARY

Analytical and result-oriented Data Scientist with 5+ years of experience in developing ML and GenAI solutions across the finance domain. Skilled in predictive modeling, natural language processing, LLMs, and stakeholder collaboration. Proven ability to design AI systems that drive compliance efficiency, automate decision-making, and enhance risk detection. Passionate about applying cutting-edge techniques to solve complex business problems.

SKILLS

Python	MySQL
Machine learning	Deep learning/NLP
Classification/Clustering	TensorFlow/Keras
Predictive modelling	Statistical Analysis
Prompt Engineering/SQL Agents	CNN
LLM / LangChain, GPT-4 (ChatOpenAI)	Pandas/Numpy /Scikit-Learn
Visual Studio Code/ Jupyter	OpenAI Embeddings/VectorDB

WORK HISTORY

Data Science Executive, 07/2022 - Current

Macquarie Group – Gurgaon, India

1. Generative AI Platform for Trade Alert Explanation and Document QA

Built a GenAI-based platform to help compliance teams investigate trade alerts by combining trade data with internal policy documents.

- Developed a Streamlit app with three modules: PDF QA, SQL Trade Query, and Alert Explainer.
- Used LangChain's RetrievalQA with Chroma vector store for querying policy PDFs using LLMs.
- Integrated SQL agents to extract trade data from databases using natural language.
- Combined structured (SQL) and unstructured (PDF) data to auto-generate alert explanations.
- Reduced manual review time for compliance teams by automating contextual analysis of alerts.

2. Transactional Wall Crossing Detection using NLP

Automated detection of MNPI (material non-public information) sharing in internal communications to

enhance regulatory compliance.

- Built an NLP classification model to flag wall-crossing instances in Teams and emails.
- Used TF-IDF, Word2Vec, and LSTM for feature extraction and classification.
- Achieved strong precision-recall tradeoff by tuning models to minimize false positives.

3. Market Abuse Detection using Machine Learning

Built a daily monitoring model to detect spoofing and manipulation by traders.

- Engineered trade behavior features and used Random Forest, XGBoost for classification.
- Automated ingestion and inference pipeline for real-time abuse detection.
- Improved detection precision and supported compliance review.

4. Unsupervised Market Abuse Trader Segmentation

Segmented traders based on unusual trading behavior to prioritize alerts for investigation.

- Applied K-Means, DBSCAN, and Isolation Forest to detect anomalous behavior.
- Delivered clusters to the surveillance team to improve alert triage and efficiency.

System Engineer, 10/2019 - 07/2022

Tata Consultancy Services – Gurgaon

- Developed ML model to predict and reduce loan default risk, enhancing credit decision accuracy.
- Migrated Alteryx workflows to Python, improving maintainability and scalability.
- Designed and deployed Power BI dashboards for actionable business insights and trend analysis.
- Tuned machine learning models for improved accuracy, recall, and precision and reduced false positives by 30%.
- Collaborated with stakeholders to gather business requirements and translate them into analytical solutions.
- Applied statistical analysis and data mining techniques to uncover patterns and inform decisions.

EDUCATION

Bachelor of Technology: Electronics And Communications Engineering, 08/2015 - 07/2019

Bharati Vidyapeeth's College Of Engineering - New Delhi

GPA: 8.1

XII(CBSE): Science Education, 03/2014 - 03/2015

National Public School - New Delhi

GPA: 90%

X(CBSE): 03/2012 - 03/2013

National Public School - New Delhi

GPA: 9.8

CERTIFICATIONS

Machine Learning Certification, Simplilearn

Data Science Professional Certification, Simplilearn

ACCOMPLISHMENTS

- Promoted to Executive position within one year of joining Macquarie for taking ownership of projects with minimal supervision.
- Awarded with "Spot Award" for efficient stakeholder management.